

Introduction to Python

Computer Science Basics

What is a Programming Language?

- Computers only understand **binary** (0s and 1s).
- A programming language acts as a translator, allowing humans to write instructions in a readable format.
- These instructions are later compiled or interpreted into the machine code the computer executes.
- **Python** is a high-level language, meaning it's designed to be easily read and written by humans.



Evolution of Programming Languages

The evolution of programming languages is divided into several generations:

Low-Level Languages

- **Machine Code:** Strings of bits (0s and 1s) executed directly by the processor.
- **Assembly Language:** Introduced short names (mnemonic codes, e.g., ADD, MOV) instead of bits, making the code easier for humans to read, but being strictly tied to the hardware architecture.

High-Level Languages

- Appearing starting in the 1950s, these use English words (such as if, while, print) and mathematical notations.
- **Historical examples:** Fortran (1957), COBOL (1959), BASIC (1964), Pascal (1970).
- **The turning point:** The C language (created in 1972 by Dennis Ritchie) combined the efficiency of low-level languages with the clear syntax of high-level ones. Many popular modern languages derive from C, such as C++, Java, JavaScript, C#, and Python.

Translating Code

Through programming languages we write programs, or source code. Programs describe the tasks the computer will perform. Computers cannot directly understand source code written in a high-level language. For the instructions to be executed, the source code must be translated into machine code. Depending on how this translation is done, programming languages are divided into two main categories:

A. Compiled Languages

- **Mechanism:** A program called a compiler takes the entire source code, analyzes it globally, and generates a standalone executable file (e.g., an .exe file on Windows).
- **Examples:** C, C++, Pascal.
- **Advantages:** Extremely fast and optimized execution.
- **Disadvantages:** Any modification requires recompiling the entire code; the executable file is dependent on the operating system on which it was created.

B. Interpreted Languages

- **Mechanism:** A program called an interpreter reads the source code line by line, translates it on the spot, and executes it immediately.
- **Examples:** Python, JavaScript, Ruby.
- **Advantages:** High portability (the same code runs on Windows, macOS, or Linux without modifications) and ease of debugging (errors are reported immediately at the exact line where they occurred).
- **Disadvantages:** Execution speed is generally lower than in the case of compiled languages.

The Evolution of "Hello, World!"

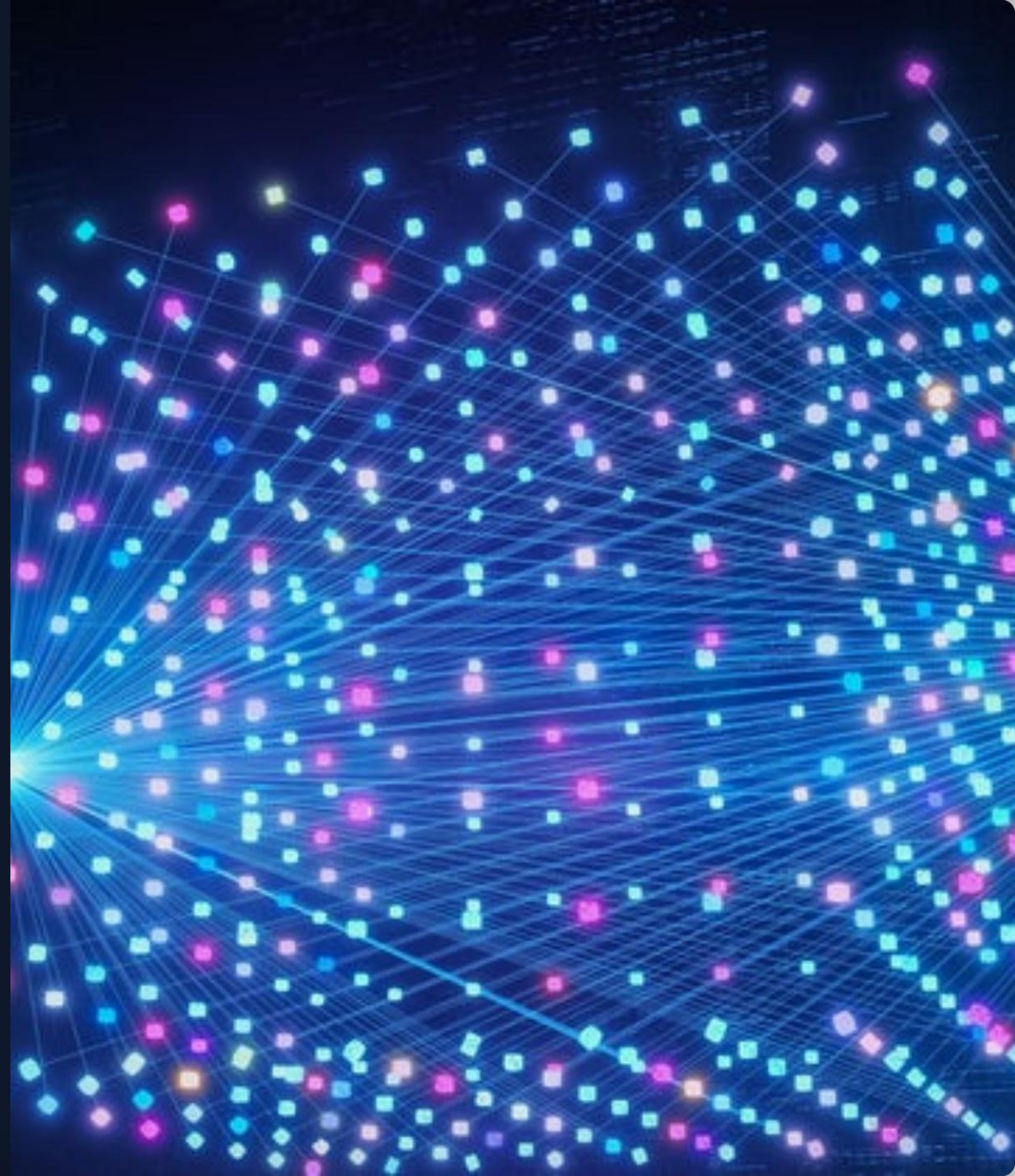
Language	Implementation
Assembly (x86)	<pre>mov ah, 09h lea dx, msg int 21h</pre>
COBOL	<pre>DISPLAY "Hello, world".</pre>
Pascal	<pre>WriteLn('Hello, world');</pre>
C	<pre>printf("Hello, world\n");</pre>
Java	<pre>System.out.println("Hello, world");</pre>
C#	<pre>Console.WriteLine("Hello, world");</pre>
JavaScript	<pre>console.log("Hello, world");</pre>
Python	<pre>print("Hello, world")</pre>

What is a **Variable**?

Think of it as a labeled box.

A variable is a named location in the computer's memory used to store data.

Just like you put a label on a moving box so you know what's inside, a programmer names a variable to easily retrieve and use its value later.



Fundamental Data Types



int

Integer: Whole numbers without a decimal point. Example: 2026, -15.



float

Floating Point: Numbers containing a decimal point. Example: 9.85, 3.14.



str

String: Text or characters enclosed in quotes. Example: "Alex", "Hello!".



bool

Boolean: Logical values of truth. Can only be True or False.

Live Code Demos

Let's write our first scripts.

Demo 1: The `print()` Function

The `print()` function is how Python communicates with the user by displaying information on the screen.

- Use quotes for displaying raw text (strings).
- Numbers do not need quotes.
- You can combine multiple items by separating them with commas.

```
# Display a simple message (string)
print("Hello! Welcome to your first CS class!")

# Display a number
print(2026)

# Display text and numbers combined
print("The current year is:", 2026)

# Using special characters (\n for new line)
print("Line 1\nLine 2")
```

Demo 2: Variables & Types

In Python, you don't need to explicitly declare a variable's type. The language figures it out dynamically based on the value you assign.

The `type()` function is incredibly useful for inspecting what kind of data is currently inside a variable.

```
# Declaring and initializing variables
age = 15 # int (integer)
gpa = 3.85 # float (decimal)
student_name = "Alex" # str (string)
is_present = True # bool (boolean)

# Displaying variables
print("Student", student_name, "is", age, "years old.")

# Revealing the data type
print("The type of 'age' is:", type(age))
```

Demo 3: Input & Casting

The `input()` function allows the program to ask the user for data.

Crucial Rule: `input()` always returns data as a **String** (text), even if the user types a number.

To do math, we must convert (or "cast") that string into a number using `int()` or `float()`.

```
# Reading text
name = input("What is your name? ")

# Reading an integer (requires casting)
birth_year = int(input("What year were you born?
"))

# Processing
est_age = 2026 - birth_year

# Output
print("Hello,", name, "!")
print("You are approx.", est_age, "years old.")
```

Demo 4: Basic Arithmetic

Python easily handles all standard mathematical operations.

- **Real Division (/)** always results in a float.
- **Integer Division (//)** gives you the quotient (how many whole times a number fits).
- **Modulo (%)** gives you the remainder of an integer division.

```
a = 15
b = 4

print("Add:", a + b) # 19
print("Sub:", a - b) # 11
print("Mult:", a * b) # 60
print("Div:", a / b) # 3.75

# Special algorithm operators
print("Int Div:", a // b) # 3 (Quotient)
print("Modulo:", a % b) # 3 (Remainder)
print("Power:", a ** b) # 50625
```

Demo 5: The Taxi Problem

A taxi charges a base fare of $\$b$ and $\$a$ for each mile driven.

If a client paid a total of $\$c$, how many miles did they travel?

$$a \cdot x + b = c \Rightarrow x = \frac{c - b}{a}$$



Solving the Taxi Problem

IPO Architecture

- Input: Read the base fare, rate per mile, and total cost as floats (decimals).
- Processing: Subtract the base fare to isolate the distance cost, then divide by the rate per mile.
- Output: Print the final calculated miles.

```
# 1. INPUT
base = float(input("Base fare ($): "))
rate = float(input("Rate per mile ($): "))
total = float(input("Total cost ($): "))

# 2. PROCESSING
dist_cost = total - base
miles = dist_cost / rate

# 3. OUTPUT
print("The client traveled:", miles, "miles.")
```

Practical Exercises (Part 1)

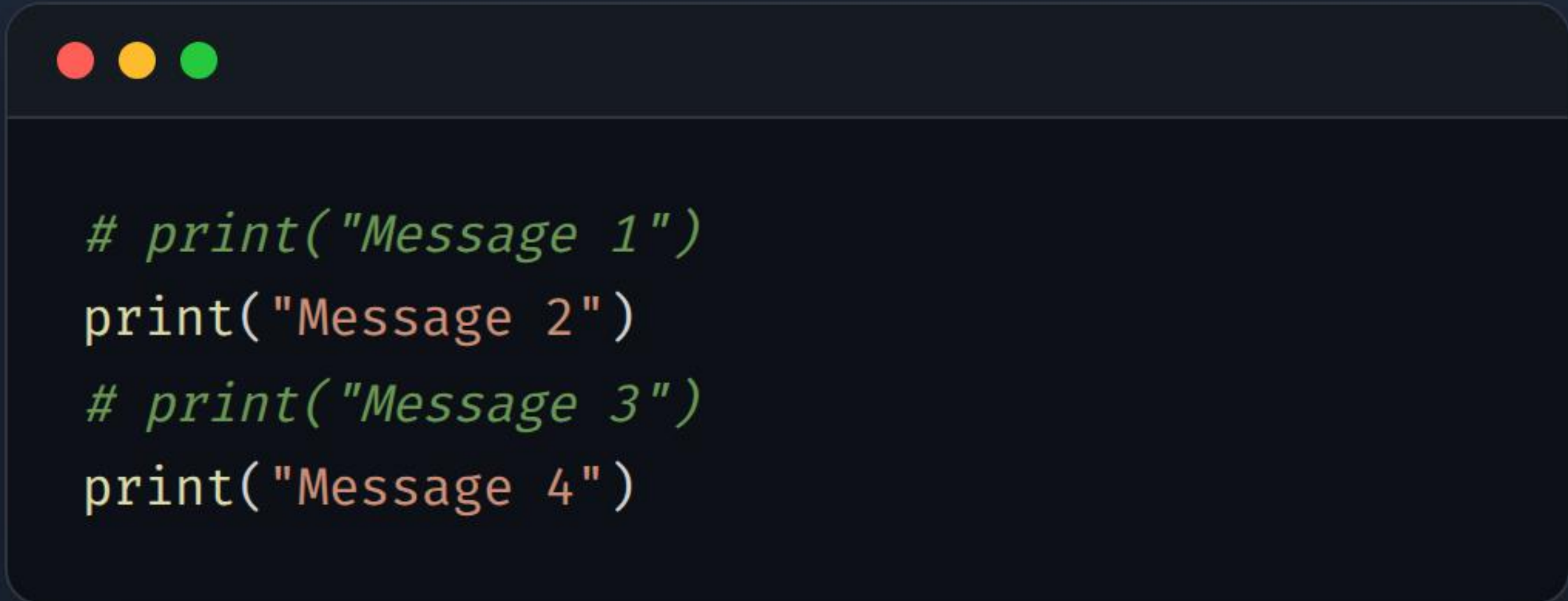
Exercise 1 (Multiple Choice)

What is the role of the # character in Python code?

1. Stops program execution.
2. Marks the beginning of a comment ignored by the interpreter.
3. Used to display numbers on the screen.
4. Compiles the file into an .exe format.

Exercise 2 (Code Analysis)

Analyze the code below. What will it display on the screen when executed?



```
# print("Message 1")  
print("Message 2")  
# print("Message 3")  
print("Message 4")
```

Practical Exercises (Part 2)

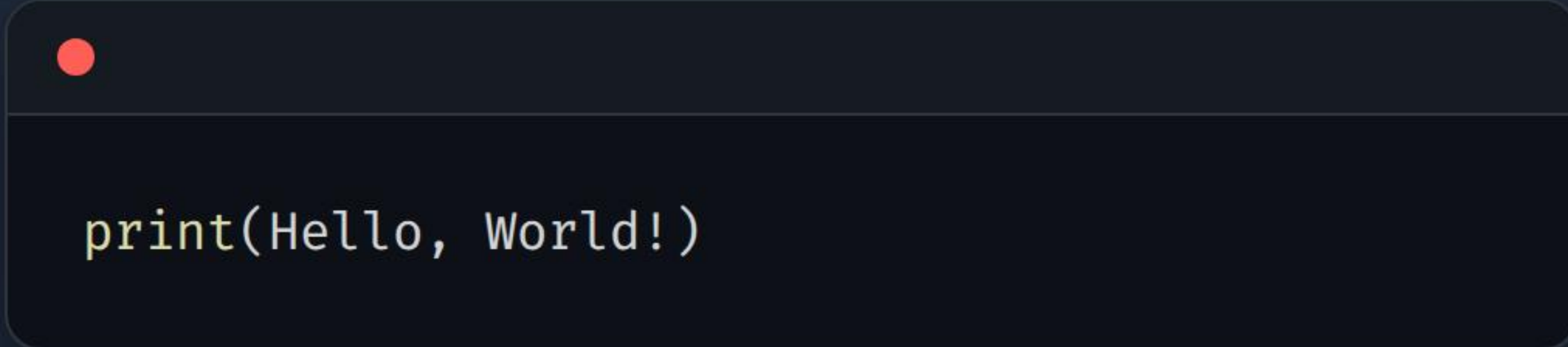
Exercise 3 (Application)

Open your IDE and write a program that:

- Contains a comment on the first line with your name and grade.
- Displays on separate lines:
 - The name of your high school.
 - Your major/specialization (e.g., Math-CS).
 - A sentence about what you want to learn in this class.

Exercise 4 (Debugging Challenge)

A student wrote this program, but received an error:



```
print(Hello, World!)
```

- Identify the mistake. Why does Python throw an error here?
- Write the corrected version of the code.

Questions?

Up Next: Making decisions with **if / else** statements.

Thank you for participating!

Image Sources



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